Xiaomi Note3 Level 3 Maintenance Guide V01

Technical support internal document control: TSIMMPC8 Xiaomi Note3 Level 3 Maintenance Guidance V01

Scope of application:
Change history of analysis center, motherboard and whole machine maintenance factory:

First edition 2017/11/7

1. Introduction of basic information

1.1 Product Overview

Processor and memory:

全系标配 6GB
大内存

大内存带来更流畅的用户体验。最高可选 128GB 大容量，存储更多游戏、视频和图片。

高通骁龙660
高端处理器

14 纳米工艺，最高主频可达 2.2GHz，处理速度更快，功耗却更低。

3500mAh 大电池
支持快充3.0

大屏大电池，为你提供了更持久的使用时间，同时支持快充3.0，半小时充电 50%。

多功能NFC
手机支付、刷公交卡

支持60座城市出行，覆盖国内全部主流银行，可在全球超过10个国家消费。

红外遥控
手机遥控家中电器

累计支持 13 种类型，多达 3 万多个型号的电器设备。

可升级至MIUI 9
快如闪电的手机系统

快如闪电的操作系统，还有一步直达的传送门、便捷的信息助手和高效的照片查找功能。

C

Zoom double camera post camera:

12MP wide-angle lens, 4-axis optical anti-shake
12MP telephoto lens, portrait
lens front camera:
16M, Beauty 4.0, Face
Unlocked. All Netcom 4.0:
The main card and the secondary card support the 4G network of China Mobile, China Unicom and Telecom and can be inserted blindly. When the main card is 4G of the three operators, the secondary card supports Unicom 3G, and does not support the simultaneous use of two telecom cards.

Network frequency band:

支持移动、联通、电信4G+ / 4G / 3G / 2G

详细网络频段

FDD-LTE (频段 B1，B3，B5，B7，B8)

TDD-LTE (频段 B34, B38，B39, B40, B41)

TD-SCDMA (频段 B34，B39)

WCDMA (频段 B1，B2，B5，B8)

GSM (频段 B2，B3，B5，B8)

CDMA1X/EVDO (频段 BC0)

1.2 Special welding fixture for Xiaomi Note3

Material code of welding fixture: SCNC020034400

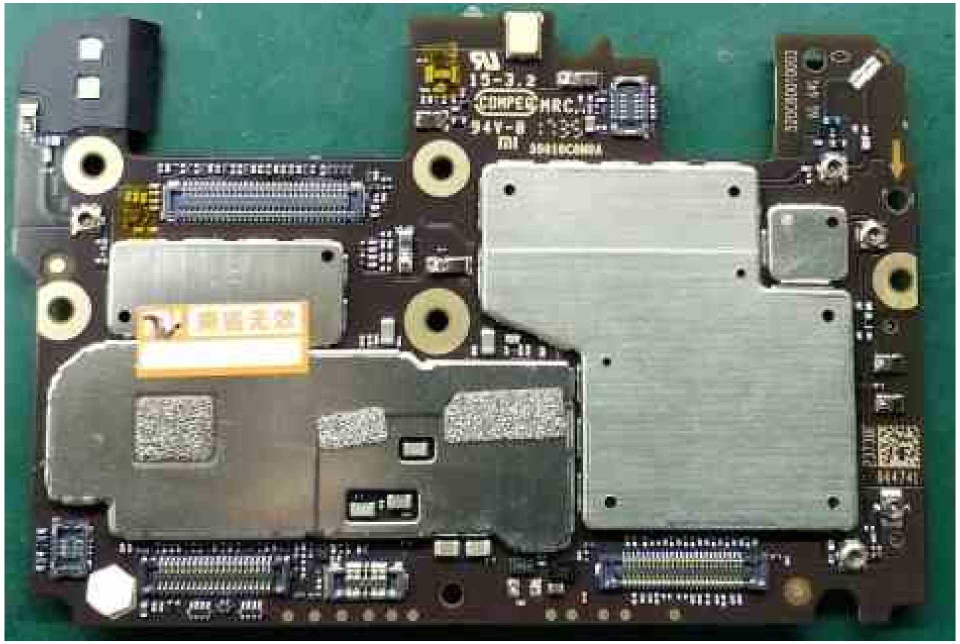
1.3 Xiaomi Note3 power supply adapter

Material code: SCNC010024000

1.4 Pasting position and specification of maintenance label

Repair label: 12mm*5mm

Paste position: Paste on SH500 and SH600 shields.



1.5 Maintenance precautions

1.5.1 Matters needing attention in motherboard welding

1. The TOP surface U1515 of the motherboard is fragile, so pay attention to protection during maintenance.

...e the heat dissipation glue on the shield cover and the chip first, and then reapply the heat dissipation glue to buckle back to the shield cover after the repair is completed.

1.5.2 Matters needing attention in function testing

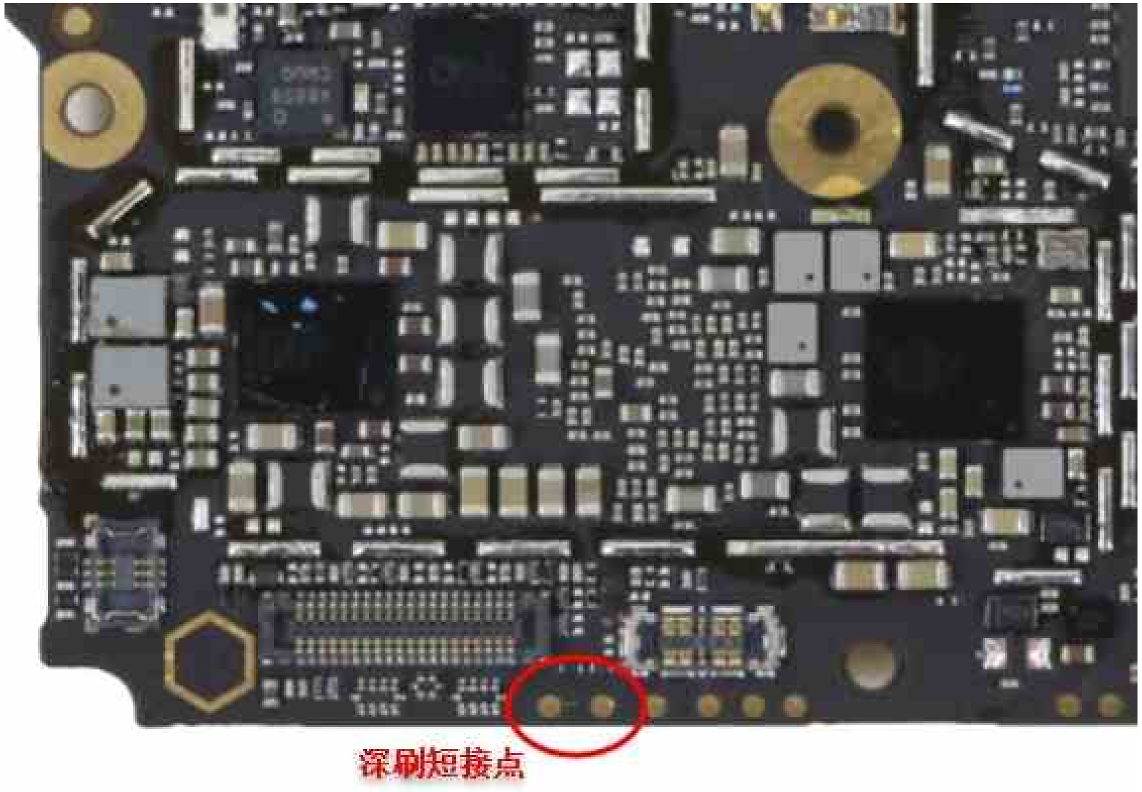
1. The Type-C to AUDIO interface used in Xiaomi Note3 headphones needs to be inserted twice before and after testing, otherwise the test will fail (headphone adapter material number:
450000026106).
2. Calibration of acceleration sensor and gyroscope sensor of Xiaomi Note3 needs to be selected in CIT auxiliary tools.

1.5.3 CPU and EMMC Replacement Rules

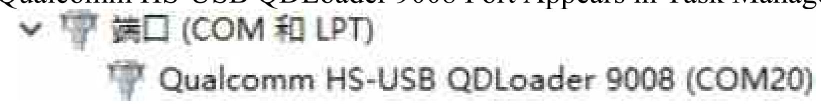
The DDR of Xiaomi Note3 is integrated with EMMC, and the CPU is bound with EMMC. EMMC can be replaced separately, but to replace CPU, EMMC must be replaced. If key is not bound, only CPU needs to be replaced.

1.6 MiFlash brush

- Shut down state, make the mobile phone enter deep brush mode (identify to port 9008) mode: short-circuit TP1204/TP1205 and connect Type-C line



- Qualcomm HS-USB QDLoader 9008 Port Appears in Task Manager



- Click on the MiFlash tool "Load Device" button, read COM port, click on the "Brush Machine" button, start brushing machine.

Erase NV factory package:

- FCK128R-C8_eraseNV

Retain NV Factory Package:

- FCK128R-C8_KeepNV

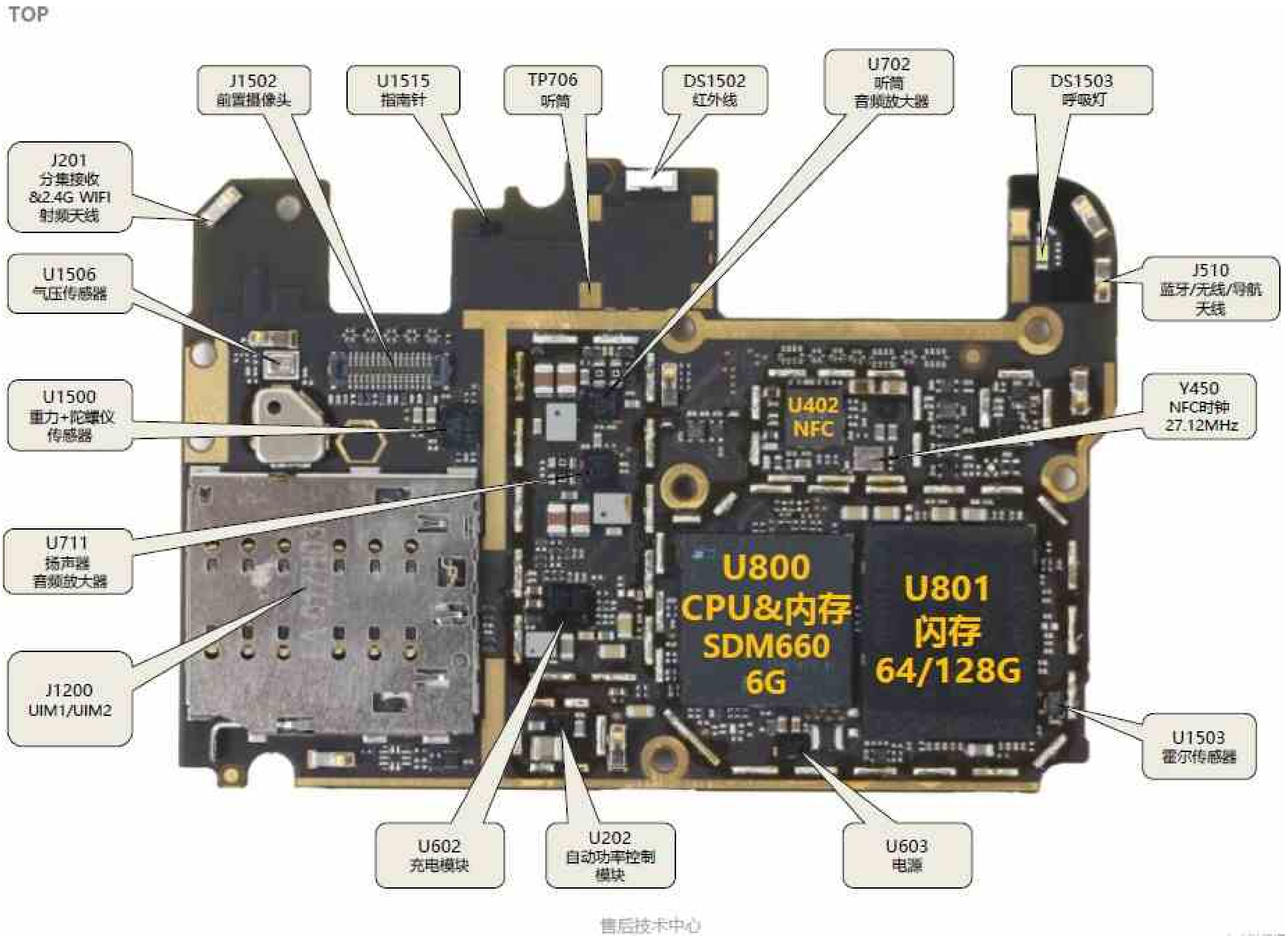
1.7 RF calibration test correlation

- Deep brush factory package (FCK128R-C8_eraseNV), written to FSN with DT tool.
- Open the calibration software, and connect the main antenna and the auxiliary receiving RF line when the mobile phone is turned on. Connect the
- USB cable, and the mobile phone starts RF calibration.
- After calibration, if you want to test the signal function, you need to use DT tool to restart the mobile phone, otherwise there is no signal, and then brush MIUI software for function test.

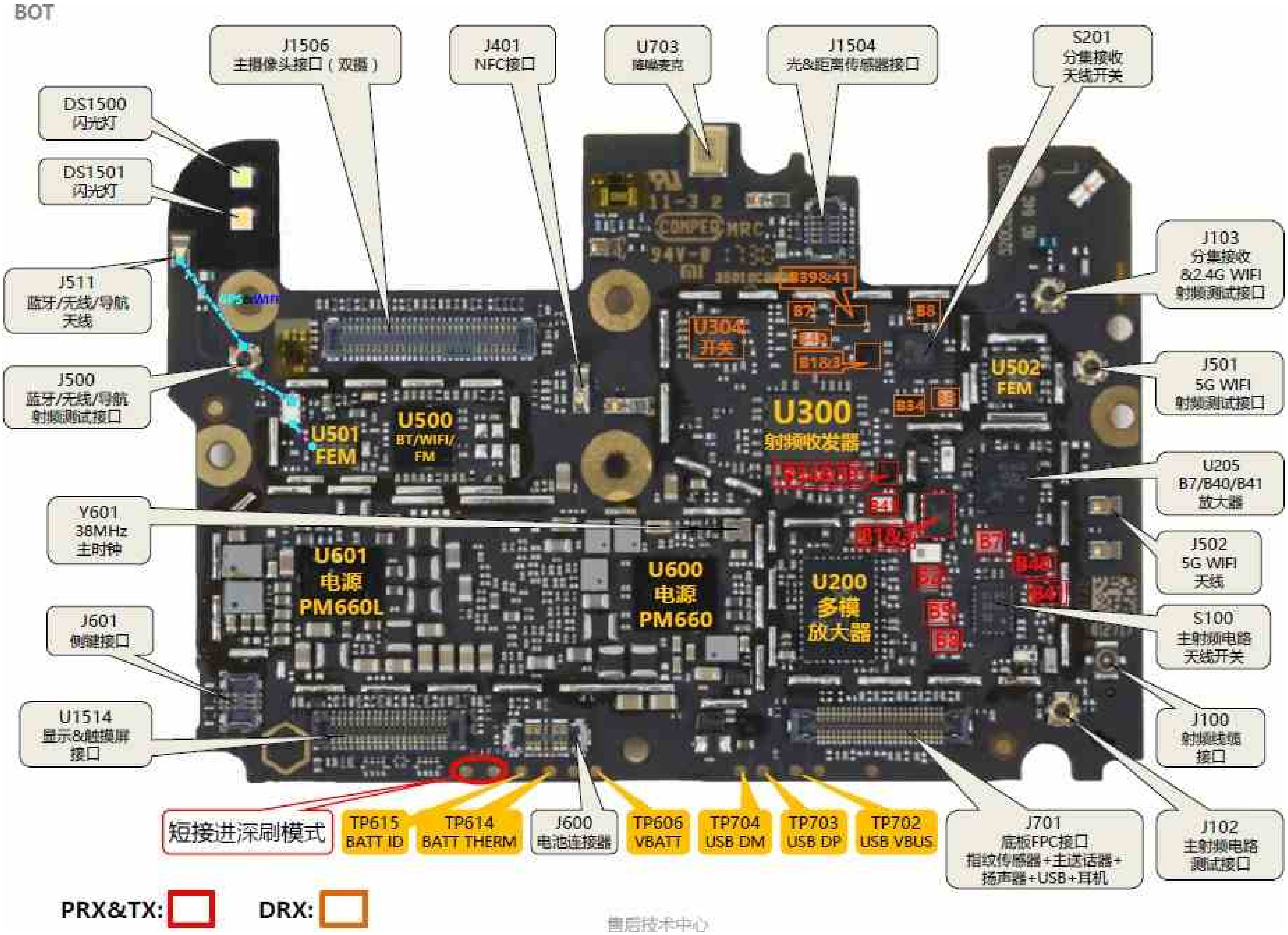
2. Brief introduction of motherboard module

2.1 Motherboard component distribution diagram

- TOP plane

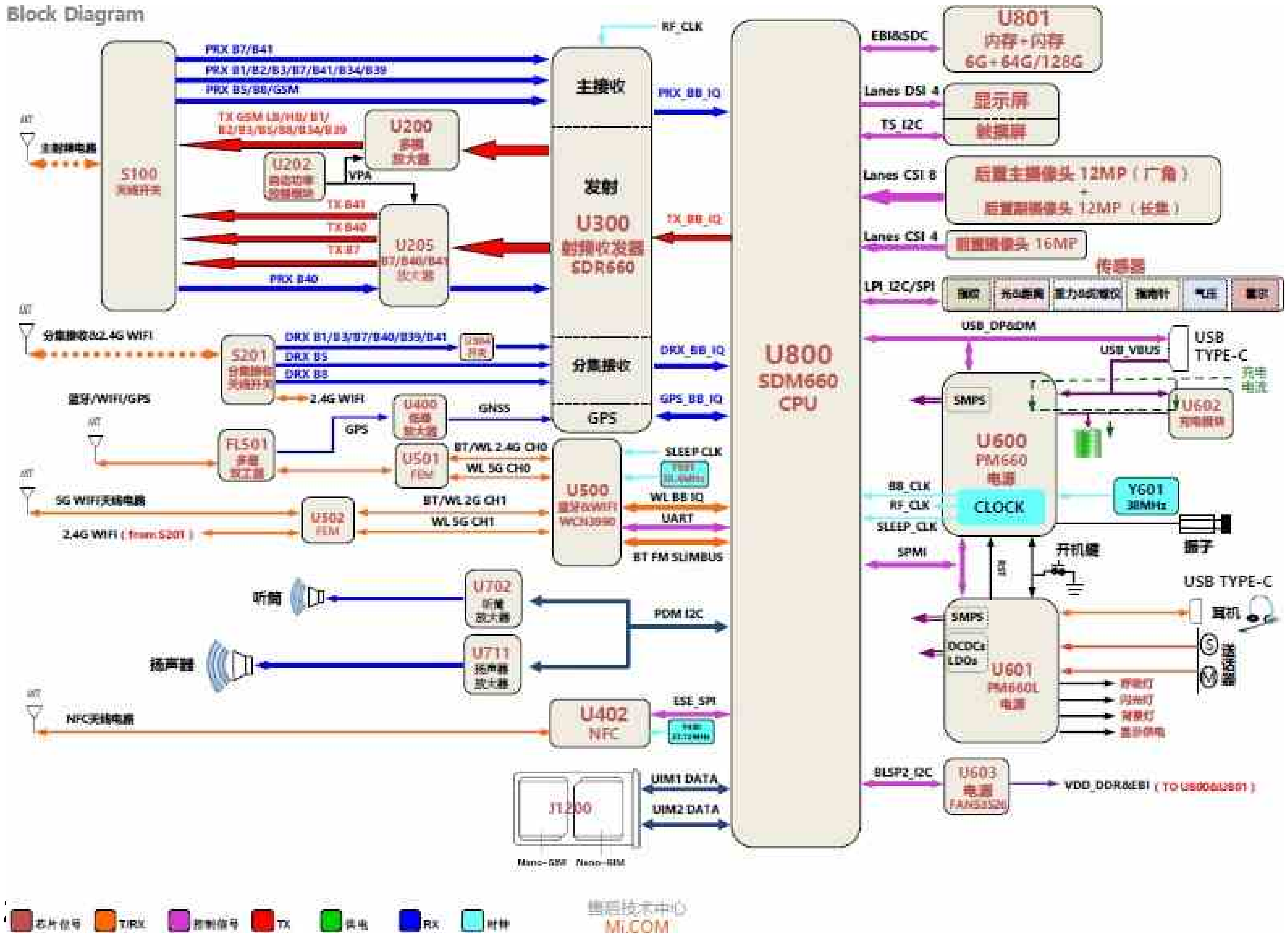

TOP
J1502
前置摄像头
U1515
指南针
TP706
听筒
DS1502
红外线
U702
听筒
音频放大器
DS1503
呼吸灯
J201
分集接收
&2.4G WIFI
射频天线
U1506
气压传感器
J510
蓝牙/无线/导航
天线
U1500
重力+陀螺仪
传感器
U402
NFC
Y450
NFC时钟
27.12MHz
U711
扬声器
音频放大器
U800
CPU&内存
SDM660
6G
U801
闪存
64/128G
J1200
UIM1/UIM2
U1503
霍尔传感器
U602
充电模块
U202
自动功率控制
模块
U603
电源
售后技术中心

- BOT surface


BOT
J1506
主摄像头接口（双摄）
J401
NFC接口
U703
降噪麦克
J1504
光&距离传感器接口
S201
分集接收
天线开关
DS1500
闪光灯
DS1501
闪光灯
J511
蓝牙/无线/导航
天线
J500
蓝牙/无线/导航
射频测试接口
J103
分集接收
&2.4G WIFI
射频测试接口
J501
5G WIFI
射频测试接口
U205
B7/B40/B41
放大器
J502
5G WIFI
天线
S100
主射频电路
天线开关
J100
射频线缆
接口
J102
主射频电路
测试接口
Y601
38MHz
主时钟
J601
侧键接口
U1514
显示&触摸屏
接口
U304
开关
U300
射频收发器
U502
FEM
U501
FEM
U500
BT/WIFI/
FM
U601
电源
PM660L
U600
电源
PM660
U200
多模
放大器
短接进深刷模式
TP615
BATT ID
TP614
BATT THERM
J600
电池连接器
TP606
VBATT
TP704
USB DM
TP703
USB DP
TP702
USB VBUS
J701
底板FPC接口
指纹传感器+主送话器+
扬声器+USB+耳机
PRX&TX:
DRX:
售后技术中心

2.2 Logic block diagram of Xiaomi Note3

Principle block diagram:



Boot Timing:

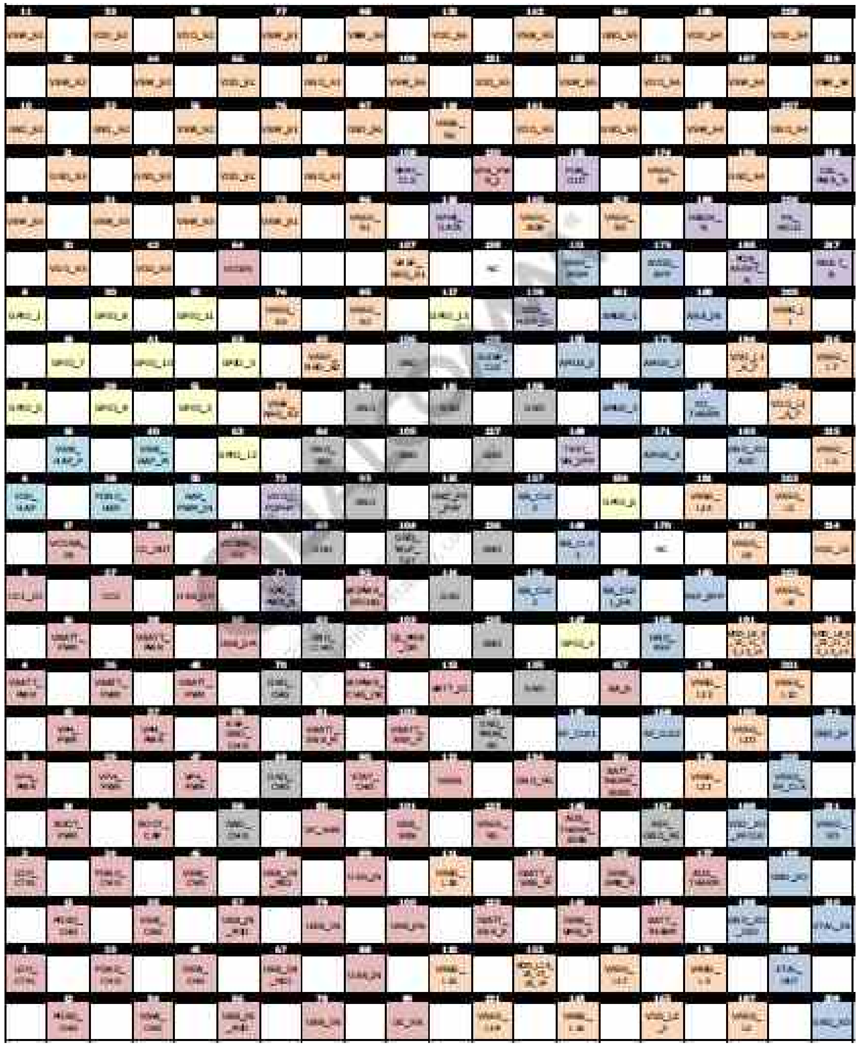
开机时序		
KEPD_PWR	C1638	1.8v
VREG_BOB	C668	3.3V
VREG_S5B	L612	0.58V
VREG_S3B_S4B	L610	0.8V
VREG_S2B	L609	1V
VREG_S4A	C618	1.8V
RFCLK	R804	38.4MHZ
VREG_S5A	C622	1.3V
VREG_L13A	C644	1.8V
VREG_L10A	C844	1.8V
VREG_L9A	C646	1.8V
VREG_L6A	C641	1.3V
VREG_L19A	C654	3.3V
VREG_L11A	C1107	1.8V
VREF_MSM_HVPAD	C837	1.25V
VREG_L10B	c1833	0.9V
VREG_L9B	C687	0.87V
VREG_L1B	C855	0.9V
VREG_L7B	C686	3.1V
VREG_S1B	L608	1.1V
VREG_L4B	C805	2.95V
SLEEP CLK	TP501	32KHz
VREG_S1A	C884	0.8V
VREG_L14A	C1596	1.8V
VREG_L1A	C637	1.25V
PON_RESET	R809	1.8V
PS_HOLD	TP600	1.8V

CPU (U800) power supply signal and measurement point, yellow part in startup timing signal:

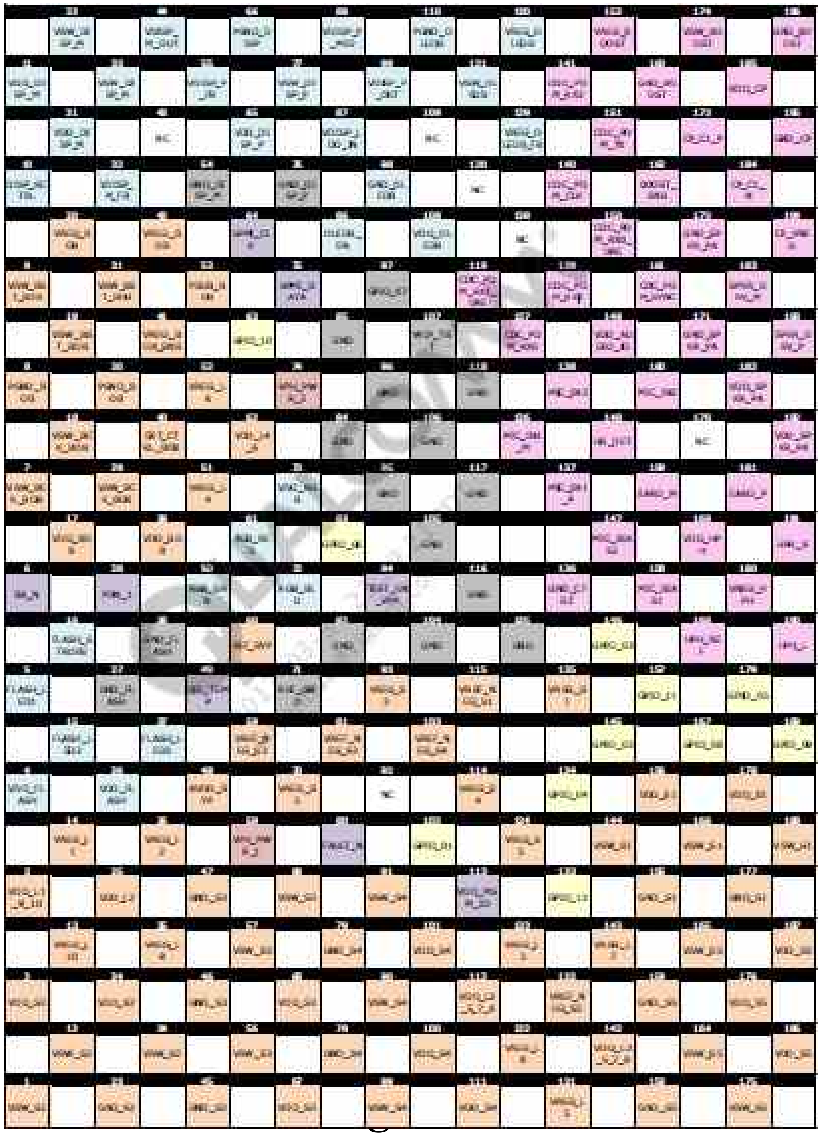
CPU主要供电		
信号	测量值	测量点
VREG_S5B_0P915	0.915	L612
VREG_L10B_0P915	0.915	c1833
VREF_MSM_HVPAD	1.25	C837
VREG_S1A_0P87	0.87	C884
VREG_S3B_S4B_0P87	0.87	L610
VREG_L9B_0P87	0.87	C687
VREG_S1B_1P125	1.125	L608
VREG_L13A_1P8	1.8	C644
VREG_L10A_1P8	1.8	C844
VREG_L1A_1P225	1.225	C637
VREG_L7B_3P125	3.125	C686
VREG_L6A_1P3	1.3	C641
VREG_L9A_1P8	1.8	C646
VREG_L1B_0P925	0.925	C855
VREG_LP4x_0P6	0.6	R824
VREG_S6A_0P87	0.87	C874
VSNS_VDDIO_EBIx	0.6	R824
VREG_S2A_S3A_0P87	0.87	C890
VREG_L5A_0P848	0.848	C640

2.4 Bitmap of main IC

PM660:



PM660L:



3.1 Switch-on and shutdown faults

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. In the specific measurement, we should measure in turn according to the sequence of starting up.

3.1.1 Analysis of

maintenance

without startup at

constant current;

1. Judge whether the motherboard is boot-up high current, if it is boot-up high current (boot-up current is greater than good motherboard in the same state), measure whether the output voltage of power supply (U600, U601) is positive

Often.

2. Software upgrade, troubleshooting software.
3. If the error is reported about 8s, measure whether the working condition of U801 is normal (the damage ratio of U801 is large in actual maintenance).

编号	设备	进度	时间	状态		结果
2	COM20		9s	fla	\C8\software\FCK128P-C8_keepNV\images\dummy.img	

Maintenance Case 1

Failure Phenomenon:
100mA Maintenance
Failure Component: U801

Maintenance analysis: Constant current does not start, brush can recognize mouth, 8s error, replace U801 repair

Maintenance Case 2

Failure phenomenon: 100-
130mA maintenance failure
component: software upgrade

Maintenance analysis: Constant current does not start up (the current jumps 270mA and falls back to 140mA during startup), and the NV factory package is reserved for repair.

3.1.2 **Analysis of**

maintenance without

maintenance of startup

current;

1. Software upgrade, troubleshooting software.
2. Measure whether the startup timing signal is normal and whether there is short circuit.
3. Measure whether the power supply of U600 and U800 is normal.

Maintenance Case 1

Fault phenomenon: 50mA
does not maintain faulty
component: U600

Maintenance analysis: 50mA is not maintained, the startup timing is measured, and the VREG_L11A/L8A/S2A voltage output by U600 is not available. Replace U600 for repair.

Maintenance Case 2

Fault phenomenon: 140mA
does not maintain faulty
component: C658

Maintenance analysis: Press the startup button, the startup current will jump to 140mA, then fall back to 10mA constant current, the brush will pass, measure the startup timing, VPH_PWR will be pulled down to 0.5 V, measure U600 peripheral components, find VREG_FG short circuit, replace C658 for repair.

Maintenance Case 3

Fault phenomenon: 140mA
does not maintain faulty
component: U800

Maintenance analysis: the startup current jumps from 50mA to 140mA, and returns to zero after releasing the startup key. Measure the startup timing, there is no PS_HOLD, and it is invalid to replace U600. Replace U800 for repair.

3.1.3 Analysis

ideas of non-

current

maintenance on

startup;

1. Check whether the appearance of J600 and J601 is damaged. If the interface is normal, measure whether the ground value of J600 is normal (BAT_ID, BAT_THERM, VBATT).
2. Power-on measurement VPH_PWR output is normal, PHONE_ON_N (C1638) 1.8 V voltage is normal.
3. Measure whether the startup timing signal is normal.

Maintenance case column 1

Fault Phenomenon: No
Current Fault
Component: L607

Maintenance analysis: There is no current when pressing the startup key or inserting the charger. Measure the startup timing, and there is no VREG_BOB. Further measurement shows that L607 is broken, and repair it after replacement.

3.1.4 Leakage does not start up

Maintenance analysis ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.
2. Measure the value of VBATT and VPH_PWR to ground, whether the voltage is normal, if normal, power up to find the heating element.
3. Measure whether the output signal circuit of power supply (U601, U600) is normal.

Maintenance Case 1
Fault Phenomenon:
Leakage 100mA Fault
Components: U600, U602

Maintenance analysis: It was found that VBATT and VPH_PWR were connected in series. After replacing U600, it still leaked 100mA. It was further judged that VPH_PWR leaked electricity, and U602 was replaced for repair.

Maintenance Case 2
Fault Phenomenon:
Leakage 1A Fault
Component: U600

Maintenance analysis: It is found that the resistance of VPH_PWR to ground is small, L606 is removed and judged as the back end. When the back end is powered on, U600 heats up obviously, so replace U600 for repair.

Maintenance Case 3
Fault phenomenon:
leakage 10mA Fault
component: U600

Maintenance analysis: The power leakage is 10mA, and the mobile phone can be turned on normally, but in the dormant state, it still leaks 10mA (normally in the dormant state, the current will return to zero). Replace U600 for repair.

Maintenance Case 4
Fault Phenomenon:
Leakage 900mA Fault
Component: U600

Maintenance analysis: Measure the leakage of VPH_PWR, the resistance value is more than 100, the normal value is about 350, U600 is obviously hot, and repair it after replacement.

Maintenance Case 5
Fault phenomenon:
leakage 180mA Fault
component: U600/U800
Maintenance analysis: VPH_PWR leaks electricity, and its resistance is low. After replacing U600, the leakage is repaired, but it still does not start up,

and 140mA is not maintained. Measurement shows that VREG_L10A is short-circuited, so replace U800 for repair.

Maintenance Case 6

Fault phenomenon: leakage 3A

Faulty components: U711, C771

Maintenance analysis: Power-on leakage for 3A, measure short circuit of VPH_PWR, remove L606, judge that it is short circuit at the back end, and there is no obvious heating element. Finally, after removing L702, VPH_PWR is no longer short circuit. After replacing U711, power-on leakage still occurs, and U711 is obviously hot. Measurement finds short circuit of C771, and repair it after removing and replacing C771.

Maintenance Case 7

Fault Phenomenon:
Leakage 3A Fault
Component: C676

Maintenance analysis: Power-on leakage 3A, judge VPH_PWR short circuit, power-on alone, find that U601 periphery is obviously hot, try to remove the capacitor, judge C676 caused.

Maintenance Case 8

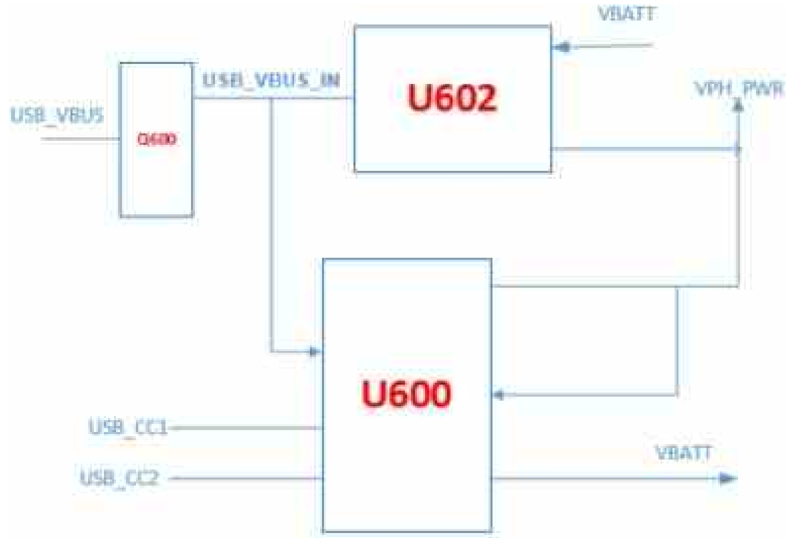
Fault phenomenon:
leakage 500mA Fault
component: U501

Maintenance analysis: Power-on leakage 500mA, VPH_PWR short circuit, power-on alone found that U501 was obviously hot, R512 was removed and judged to be U501 problem, which was repaired after replacement.

3.2 Charging function failure

The USB interface of Xiaomi Note3 adopts TYPE-C interface, which supports positive and negative insertion. It is recognized by USB_CC1/CC2. U600 and U602 are used for parallel charging, and VBUS enters U600 and U602 through an overvoltage protection IC (Q600).

Charging schematic diagram:



Details of resistance value of each PIN pin of daughter board interface J701:

		SUB BOB J701																										
		1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49		
		GND	350	350	GND	420	420	420	420	GND	430	0	320	370	430	420	420	510	510	420	470	430	GND	650	640	530		
G1	530																										GND	G2
		GND	770	580	580	580	580	580	580	770	GND	720	580	870	GND	630	630	GND	480	GND	540	GND	590	590	GND	800		
		2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50		

Maintenance analysis ideas:

1.Software upgrade, troubleshooting software.

2.Check whether the appearance of J701 is normal, measure whether the four groups of signals USB3_VBUS, USB3_HS_DP, USB3_HS_DM and USB3_HS_ID are normal to ground with multimeter diode file, measure whether the VBUS voltage value at both ends of Q600 input and output is 5V, and focus on checking whether there is virtual welding in R601 and R602.

3.Measure whether there is short circuit in USB_CC1 and USB_CC2 resistance. If it is abnormal, focus on checking U600 and CR707.

3.Measure whether the VBATT, BAT_ID and BAT_THERM of J600 are normal to ground.

4.Ensure that the above signals are normal, consider replacing U600/U602

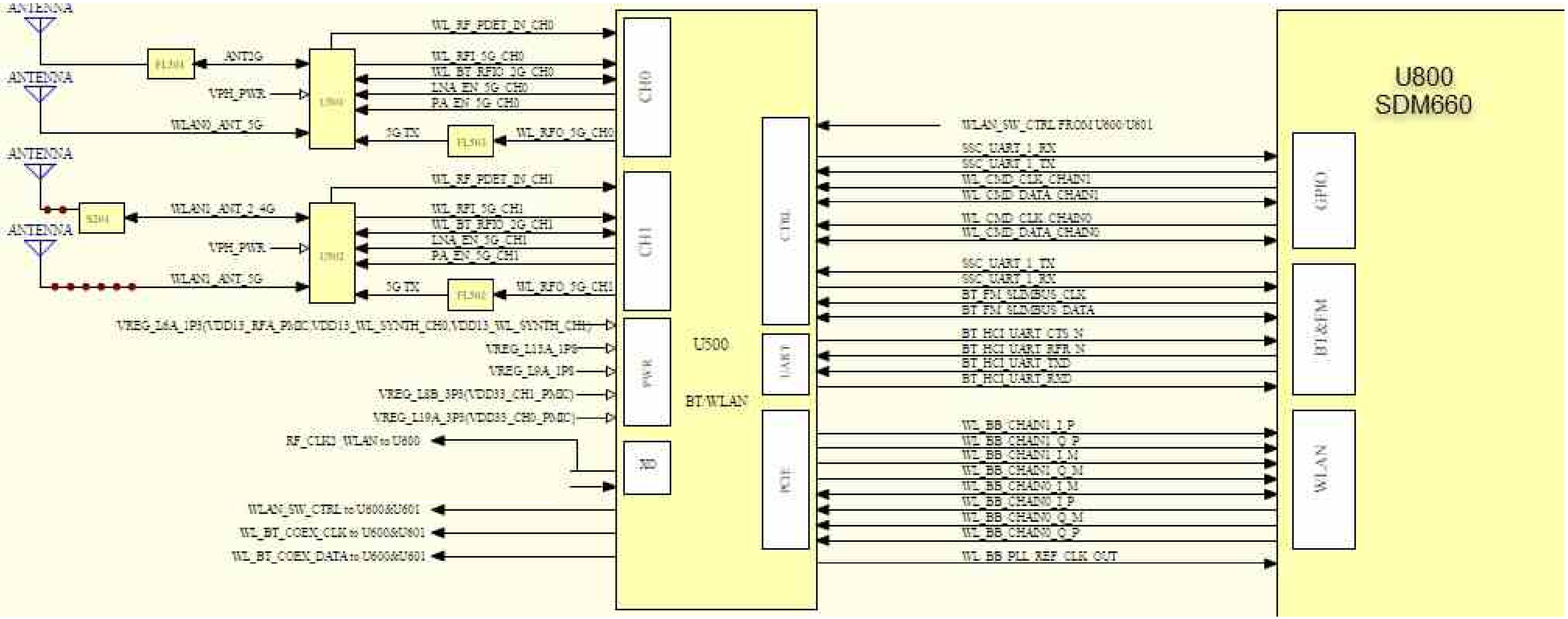
Maintenance Case 1

Fault Phenomenon:
No Charge Faulty
Component: R602

Maintenance analysis: Measure VBUS, USB_DP/DM resistance is normal, measure VBUS 5V at TP702, measure Q600 output (C660) without voltage, replace Q600 unresolved, measure peripheral devices, find R602 virtual welding, replace R602 for repair.

3.3 WIFI/BT function failure

Principle block diagram:



U500 meter:

FM、BT、WIFI测量表		
Symbol	测量值	测量点
VDD13_RFA_PMIC（VREG_L6A_1P3)	1.3V	C528
VREG_L13A_1P8	1.8V	C533
VREG_L9A_1P8	1.8V	C535
VDD33_CH0_PMIC(VREG_L19A_3P3)	3.3V	C542
VDD33_CH1_PMIC(VREG_L8B_3P3)	3.3V	C538
VDD13_WL_SYNTH_CH0（VREG_L6A_1P3)	1.3V	C524
VDD13_WL_SYNTH_CH1（VREG_L6A_1P3)	1.3V	C525
RF_CLK2_WLAN	38.4MHZ(打开wifi)	C501

Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether CLK is normal (38.4 MHz measured at C501).
3. Measure whether the power supply, clock and enable signals of U500 are normal.
4. Remove U500 and measure whether the bus between U800 is normal. If it is normal, replace U500.
5. Replace U800.

Maintenance Case 1

Fault phenomenon: WIFI
can't open faulty
component: U800

Maintenance analysis: WIFI can not be turned on, BT is normal, measuring U500 working conditions are normal, welding U800 fault repair.

Maintenance Case 2

Fault phenomenon: WIFI can't be
opened, BT can't be opened. Fault
component: U800

Maintenance analysis: WIFI can't be turned on, BT can't be opened, measuring U500 working conditions are normal, welding U500 is invalid, replacing U800 for fault repair.

Maintenance Case 3

Failure phenomenon: WIFI can't be opened

Faulty component: U500

Maintenance analysis: WIFI can not be turned on, BT is normal, measuring U500 working conditions are normal, welding U500 fault repair.

Maintenance Case 4

Fault phenomenon: BT can't search for hot spot fault component: U500

Maintenance analysis: WIFI is turned on normally, BT can't search for hot spots, welding U500 is invalid, replacing U500 for fault repair.

Maintenance Case 5

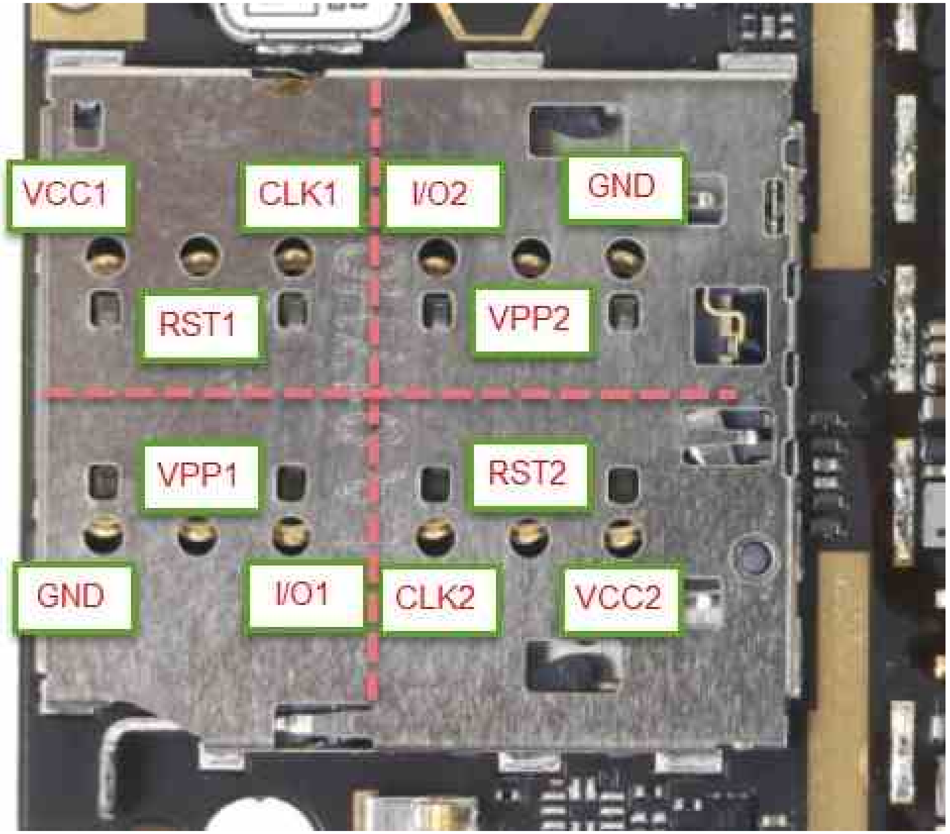
Fault phenomenon: BT can't open faulty component: U500

Maintenance analysis: BT can not be opened, measuring U500 power supply is normal, welding U500 is invalid, replacing U500 fault repair.

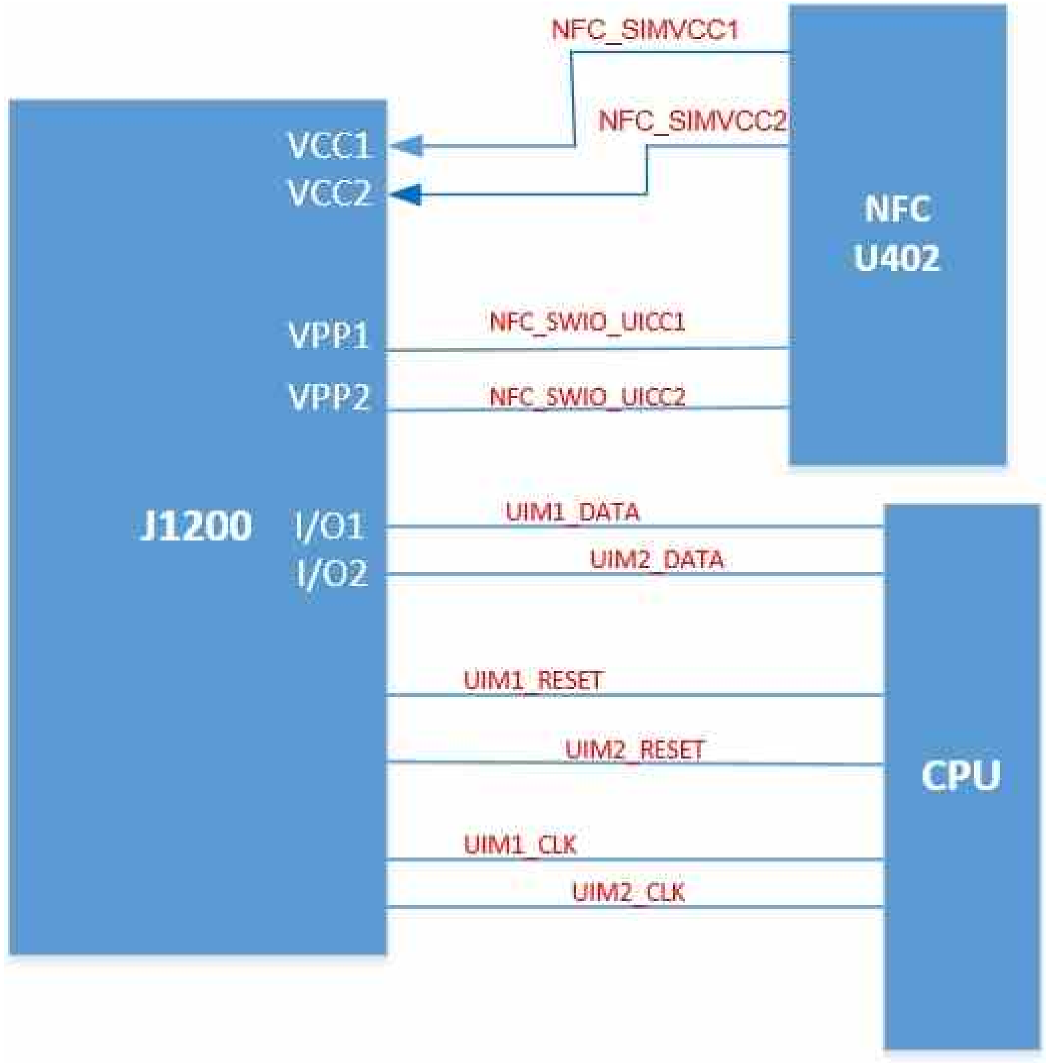
3.4 SIM card function failure

SIM card signal control chart:

SIM卡测量表		
Symbol	测量值	测量点
VCC1(NFC_SIMVCC1)	2.9V(插卡测)	C1200
VREG_L15_UIM2	2.9V(插卡测)	C1205
RST1(UIM1_RESET)	420	PIN脚
RST2(UIM2_RESET)	420	PIN脚
CLK1	420	PIN脚
CLK2	420	PIN脚
Vpp1(NFC_SWIO_UICC1)	580	PIN脚
Vpp2(NFC_SWIO_UICC2)	580	PIN脚
I/O1	420	PIN脚
I/O2	420	PIN脚



Principle block diagram of SIM card:



Maintenance analysis ideas:

1. First, check whether the baseband version of the mobile phone is normal. If the baseband information is normal, it is the SIM card function failure.
2. Check whether the SIM card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joint of SIM card holder.
3. Measure whether the SIM card aims at the ground value normally.
4. Start up and test whether the voltage of SIM card power supply, clock, reset and data signal is normal.
5. If the above signal is normal, replace U800.

Maintenance Case 1
Failure Phenomenon:
Ignorance of SIM Card
Failure Component:
J1200

Maintenance analysis: J1200 appearance is bad, and the fault is repaired after replacement.

3.5 Restart type failure

Maintenance analysis ideas:

1. Check whether the PHONE_ON_N 1.7 V voltage is pulled low.
2. Software upgrade, troubleshooting software.
3. Measure whether the I2C line is normal.
4. Measure whether there is short circuit in the output of U600 and U601, and focus on checking the power supply of U500/U300.
5. The software upgrade is invalid. Measure whether the U800 power supply and clock are normal.
6. Replace U800/U801

Maintenance Case 1

Fault phenomenon: White rice restart

Faulty component: U800/U801

Maintenance analysis: White rice restart, MIUI software upgrade invalid, deep brush erasing NV factory package invalid, welding U800/U801 troubleshooting. Maintenance Case 2
Failure phenomenon: restart
failure during startup cause:
software failure
Maintenance analysis: restart the startup process, and repair the MIUI software upgrade fault. Maintenance Case 3
Failure phenomenon: Restart
faulty component during
startup: U300

Maintenance analysis: restart the startup process, measure the output voltage of U600, find VREG_L3A voltage short circuit, disconnect R300 and judge that it is U300 fault, and repair it after replacing U300.

3.6 Signal function failure

Maintenance analysis ideas:

1. Insert SIM card to ensure normal recognition, and eliminate faults that do not recognize SIM card.
2. Software upgrade, troubleshooting software.
3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
4. Measure whether RF circuit power supply is normal, and whether RFFE (1/3/4/5/6) CLK/DATA is normal.
5. If the RF calibration is normal, there is still no signal, check whether the SIM card circuit (focusing on measuring whether the DATA signal is normal) is normal.

Maintenance Case 1
Failure phenomenon: Mobile card can't call out Failure cause: S100

Maintenance analysis: Calibrate the error GSM 900 TX, however, replace the antenna switch S100 for repair.

Maintenance Case 2
Fault phenomenon: weak signal fault cause: U300

Maintenance analysis: Signal only 1 ~ 2 grid, calibrate error FTM_X0_CAL, replace U300 repair.

3.7 Inductor function failure

Xiaomi Note3 has U1500 (G + G), U1503 (HALL), U1515 (Compass), U1506 (Press), J1504 (P-Sensor interface), distance sensors, pressure sensors and compasses share the same I2C.

Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether the working conditions of the corresponding sensors are normal.
3. Replace the corresponding sensor element.
4. Replace U800.

3.8 Crash type fault

Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.

2. Measure whether BAT_ID and BAT_THERM are normal.

3. Measure whether the working condition of U801 is normal and replace U801.

4. Measure whether the power supply of U800 is normal and replace U800.

Maintenance Case 1

Fault Phenomenon:
White Rice Fixed
Screen Fault
Component: Software
Upgrade

Maintenance analysis: White rice fixed screen, brush retained NV factory package repair.

Maintenance Case 2

Fault Phenomenon:
White Rice Fixed
Screen Fault
Component: Replace
U800

Maintenance analysis: It can start up, brush the machine to report error ping target failed, the current is maintained at 100mA, the replacement of U801 is invalid, and the replacement of U800 is repaired.

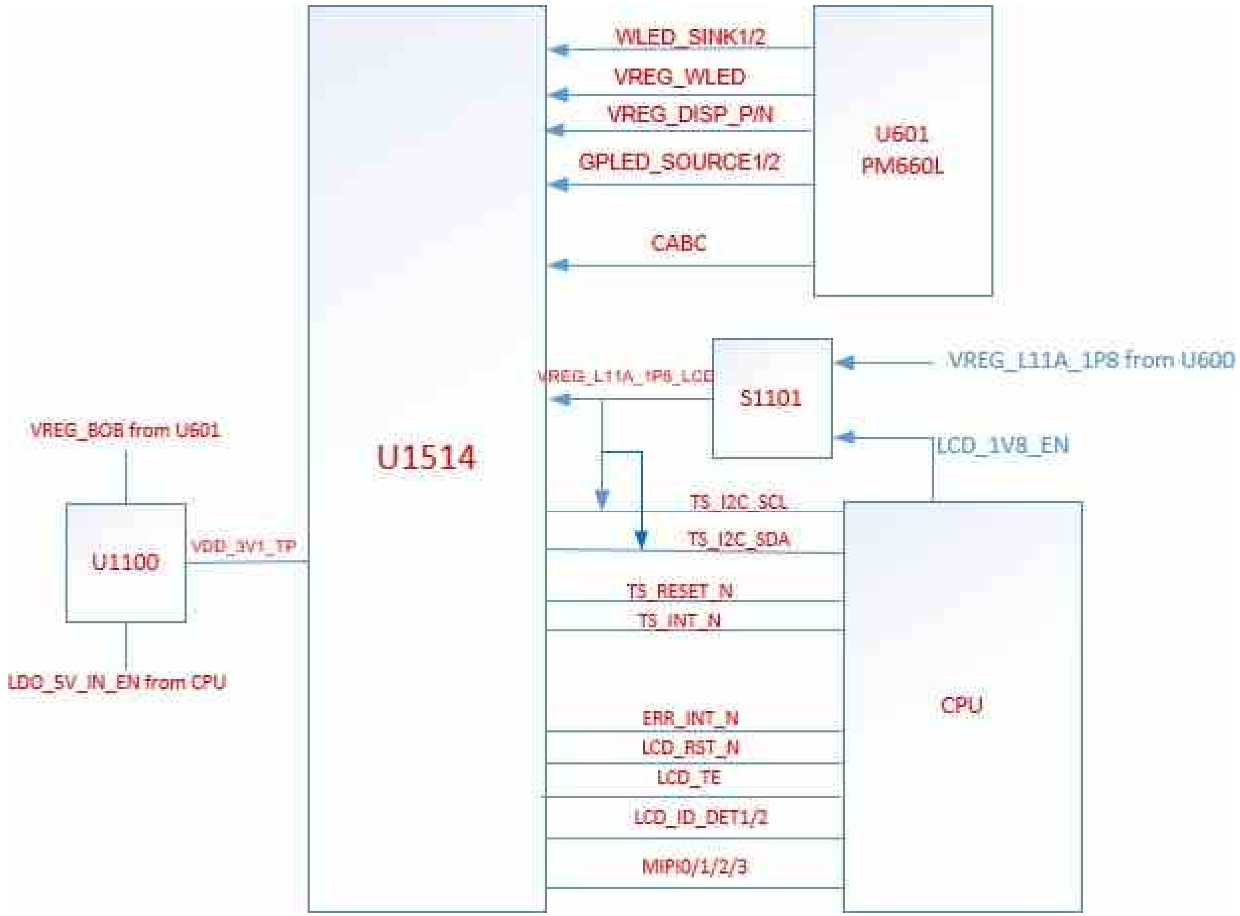
Maintenance Case 3

Fault Phenomenon:
White Rice Fixed
Screen Fault
Component: Replace
U801

Maintenance analysis: White rice fixed screen, brush function, still white rice fixed screen, measuring U801 power supply found no problem, replace U801 repair.

3.9 Display function failure

U1514 integrated LCD circuit, keyboard lamp circuit,
TP circuit. Display, TP circuit principle block
diagram:



Display measuring table

显示测量表U1514		
Symbol	测量值	测量点
VREG_DISP_P	5.5V	C1102
VREG_DISP_N	-5.5V	C1129
VREG_L11A_1P8_LCD	1.8	C1107
WLED_SINK1	0.4	C1108
WLED_SINK2	0.4	C1127
VREG_WLED	4V(息屏)，24V（亮屏）	L617
LCD_TE	420	C1106
CABA	490	R1103
MIPI_EMI_LANE(0:3)_P/N	270	FL1101/02
MIPI_EMI_CLK_P/N	270	FL1104

Display interface resistance reference

LCM CONN U1514																			
1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39
GND	270	270	GND	270	270	GND	270	270	GND	270	270	GND	270	270	GND	420	420	GND	470
610	610	GND	420	420	420	420	550	420	420	510	510	GND	490	420	==	590	GND	620	620
2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40

Maintenance analysis ideas:

1. Visually check whether U1514 and peripheral components are damaged or falsely welded.
2. Software upgrade, troubleshooting software.
3. Measure whether the ground value of each pin of U1514 is normal with multimeter diode gear.
4. If the ground value is normal, measure whether the power supply and control signals in U1514 meter are normal.
5. Replace U800.

Maintenance Case 1

Fault
Phenomenon:
Black Screen
Fault Component:
U600
Maintenance analysis: Black screen measurement DISP_N without voltage output, short circuit to ground, replace U601 fault repair. Maintenance Case 2

Fault phenomenon: black screen without display

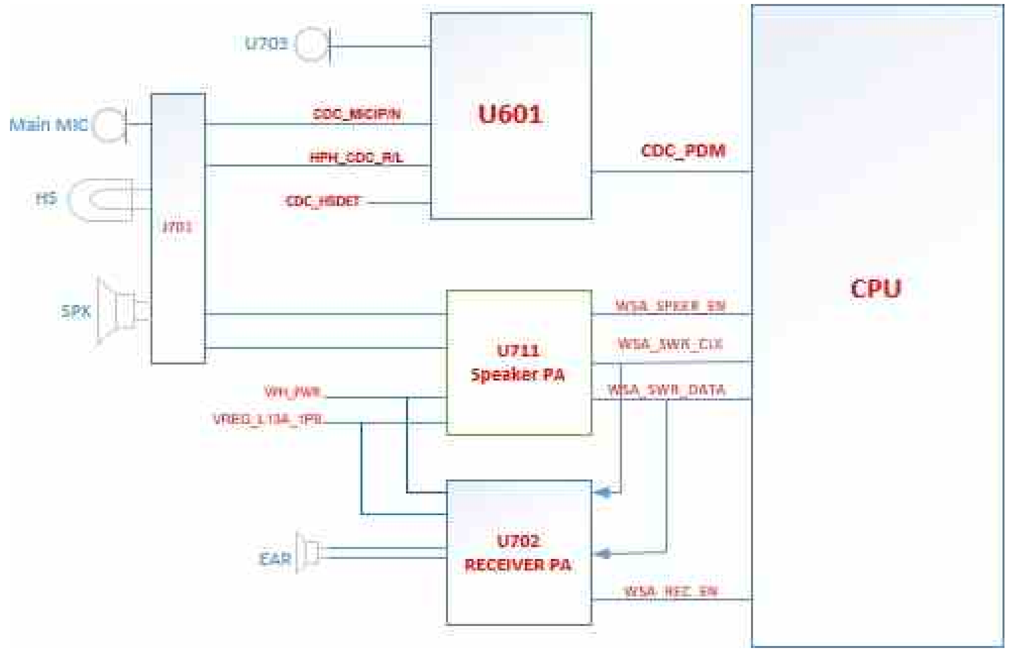
Faulty component: FL1101/1102

Maintenance analysis: MIPI_EMI_LANE0_N, MIPI_EMI_LANE2_N short circuit was found by measuring U1514 interface to ground, and the resistance value was normal after removing FL1101 and FL1102 respectively, which was repaired after replacement.

3.10 Audio function failure

The audio circuit of Xiaomi Note3 is divided into two parts:

1. The MIC and the earphone part carry out audio signal transmission through U601 and U800.
2. The earpiece and speaker have separate power amplifiers (U702/U711), and the audio signal comes from U800, which shares CLK and DATA lines. U800 controls the two power amplifiers through enable signals.
3. The earphone uses TYPE C to AUDIO interface, and the earphone detection is realized by U601, which is effective at low level. Audio signal block diagram:



3.10.1 Analysis of

loudspeaker function fault

maintenance;

1. Visually check whether J701 looks good.

2. Software upgrade, troubleshooting software.
3. Measure whether SPKR_P1 and SPKR_N1 are normal to ground with multimeter diode gear.
4. Measure whether the working conditions of U711 are normal, and replace U711 abnormally.
5. Replace U800.

Maintenance Case 1
Fault phenomenon:
loudspeaker has current sound
fault component: U711

Maintenance analysis: External current sound, replace U711 fault repair.

Maintenance Case 2
Fault Phenomenon:
Speaker Silent Fault
Component: U711

Maintenance analysis: The speaker is silent, the earpiece is normal, BOOST_OUT is found to be short-circuited by measurement, and C778/C771/C761 is still short after removal, so replace U711 for repair.

3.10.2 Analysis of

MIC Functional

Failure;

1. The CIT test determines which MIC is faulty, the main MIC, the motherboard MIC (U703)
2. Measure whether MIC power supply is normal and whether MIC_BIAS1 is normal.
3. Measure whether the primary and secondary MIC paths are normal, and if it is the main MIC, check whether the interface J701 is abnormal; If this is a secondary MIC, replace U703.
4. Replace U800.

Maintenance case
Fault phenomenon:
secondary MIC does not
send calls. Fault
component: L703

Maintenance analysis: There is no MIC_BIAS1 (normal 1.8 V) in measurement, and L703 is found to be disconnected by further measurement, which is repaired after replacement.

3.10.3 Handset function failure

Maintenance analysis ideas:

1. Is the appearance inspection normal
2. Software upgrade, troubleshooting software
3. Measure whether the receiver path is normal and whether the resistance value of TP705/6/9/10 is abnormal.
4. Measure whether the power supply and enable signal of U702 is normal, and replace U702 if it is abnormal.
5. Replace U800

Maintenance case
Fault phenomenon:
earpiece current tone
fault cause: U702

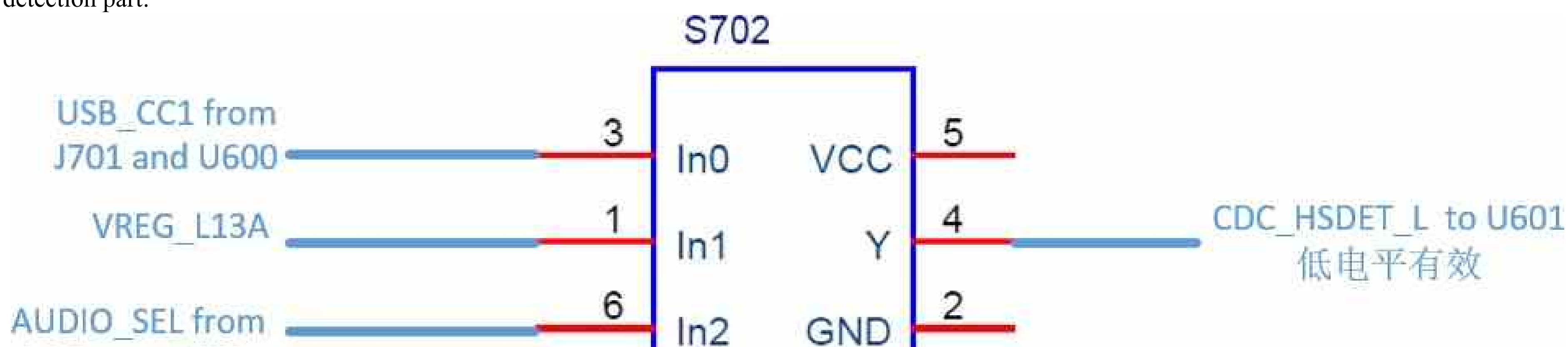
Maintenance analysis: CIT test and telephone receiver have current sound, software upgrade is invalid, no abnormality is found in measurement, replace U702 for repair.

3.10.4 Headphone

function failure

headphone part

principle:
The headphone detection is U601 detection, active low level, through USB_CC1 and AUDIO_SEL signals (these two signals come from U600), through S702 to achieve. If these two
There are abnormalities in 10
signals, so focus on checking U600.
Schematic diagram of earphone
detection part:



Maintenance analysis ideas:

1. Check whether J701 interface contacts are deformed and whether there are foreign bodies in the interface.
2. Measure whether USB_CC1 is low and whether AUDIO_SEL is high. Focus on checking whether the resistance value of USB_CC1/CC2 is short-circuited or abnormal, and replace U600 if it is abnormal.
3. Repair the headphone channel according to the principle block diagram, and replace U601 normally.
4. Replace U800.

Maintenance Case 1:

Failure phenomenon: earphone mode

Cause of failure: U600, CR707

Maintenance analysis: Under the condition of not inserting headphones, it is found that USB_CC1/CC2 is at low level (2.1 V under normal conditions) and AUDIO_SEL is at high level of 1.8 V (0V under normal conditions). USB_CC1/CC2 is short-circuited. After replacing U600, USB_CC1 is still short-circuited, and CR707 is replaced for repair.

Maintenance Case 2

Fault phenomenon: headphones are not recognized, and charging is slow. Fault causes: CR707, U600

Maintenance analysis: Measure USB_CC1 short circuit, remove CR707 no longer short circuit, still do not recognize headphones, measure USB_CC1/CC2 voltage value is low, replace U600 repair.

3.11 Camera function failure

Xiaomi Note3 rear camera adopts zoom dual camera, including wide-angle camera and telephoto camera.

Camera failures are divided into: rear main camera, rear sub-camera and front camera failures. The clock, data and I2C signals are all separate, the camera power supply comes from different voltage regulators, and the enable signal is controlled by U601.

Camera power supply signal measuring table:

相机信号测量表		
Symbol	测量值	测量点
VCM_CAM1_2P8	2.8	C1593
OIS_CAM1_2P8	2.8	C1634
DVDD_CAM1_1P1	1.1	C1595
VREG_L14A_1P8	1.8	C1596
AVDD_CAM1_2P8	2.8	C1522
AVDD_CAM3_2P8	2.8	C1597
DVDD_CAM3_1P0	1	C1527
VCM_CAM1_2P8	2.8	C1591
AVDD_CAM2_2P95	2.95	C1589
DVDD_CAM2_1P2	1.2	C1555
VREG_L14A_1P8	1.8	C1537
VREG_BOB	3.3	C1513
VREG_BOB_LDO	3.9	C1562
VREG_S5A_1P35	1.35	C1532

Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Detect J1506, J1502 and surrounding components for damage.
3. Measure whether J1506 and J1502 are normal to ground.

Main Camera J1506

1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39	41	43	45	47	49	51	53	55	57	59
GND	560	555	GND	580	580	GND	430	450	450	430	450	450	GND	560	GND	570	GND	430	560	—	550	570	430	430	GND	580	580	GND	560
780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND
2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60

Sub Camera J1502

1	3	5	7	9	11	13	15	17	19	21	23	25	27	29
GND	430	GND	GND	560	GND	430	430	GND	450	450	GND	580	GND	425
GND	780	780	GND	780	780	GND	780	780	GND	780	780	GND	780	780
2	4	6	8	10	12	14	16	18	20	22	24	26	28	30

4. Measure whether the camera power supply, clock and reset signal output are normal.
5. If the above signals are normal, replace U800.

Maintenance Case 1
Fault phenomenon: The main camera can't open the faulty component: U800

Maintenance analysis: The main camera can't be opened, the resistance of MIPI_CSI0_CLK_M is found to be infinite by measurement, and U800 is welded for repair.

Maintenance Case 2
Fault phenomenon: The main and auxiliary cameras can't open the faulty component: R1510

Maintenance analysis: The main and front cameras can't be opened. Check the public parts and find that VREG_BOB_LDO voltage is not available. Further measurement finds that R1510 is virtual welded, and repair it after replacement.

3.12 Touch screen function failure

The touch screen and display interface share U1514, and there is a separate touch screen power supply voltage regulator. Touch screen signal measurement table:

触屏测量表U1514		
Symbol	测量值	测量点
VDD_3V1_TP	3.1V	c1105
TP_INT_N	420	U1514 pin20
TP_I2C_SDA	420	U1514 pin14
TP_I2C_SCL	420	U1514 pin12
TP_RST_N	420	U1514 pin18

Maintenance analysis ideas:

1. Check and surrounding components for damage.
2. Software upgrade, troubleshooting software.
3. Measure whether the signal in the touch screen meter is normal. Measure whether the voltage value of touch screen power supply VDD_3V1_TP is normal.
4. Replace U800.

3.13 NFC function failure

NFC Signal Measurement Table:

NFC信号测量表		
Symbol	测量值	测量点
VREG_L13A_1P8	1.8V	C457
VDD_UP	3.3V	C424
NFC_ESE_SPI_CLK	400	TP410
NFC_ESE_SPI_MISO	400	TP411
NFC_ESE_SPI_MOSI	400	TP412
NFC_ESE_SPI_CS_N	400	TP413
NFC_INT_REQ	430	TP406
NFC_CLK	27.12MHZ	C472

1. Check whether J401, J402 and their circuits are deformed and damaged.
2. Software upgrade, troubleshoot software, and clear SE to verify whether it passes.
3. Measure whether the power supply, clock, reset and control signals of U402 are normal.
4. If the above signal is normal, take off U402 to measure whether the communication line with U800 is normal, and replace U402 normally.
5. Replace U800.

Maintenance Case 1

Fault phenomenon: NFC can't open faulty component: U402

Maintenance analysis: There is no measuring power supply clock, and no abnormality is found in measuring resistance. Replace U402 for repair.

3.14 USB related failures

Xiaomi Note3 adopts type C interface, supports positive and negative insertion of USB, and is realized by USB_CC1/CC2. When testing, be sure to insert USB cable forward and backward for verification.

Analysis of maintenance ideas:

1. Check the interface J701 for damage, foreign matter, etc.
2. Measure whether the resistance of VBUS and USB_HS_DP/DM is abnormal and check whether it can be charged normally.

3. Insert USB cable forward and backward, check whether it can be identified, check whether the resistance value of USB_CC1/CC2 is normal, and replace U600 if it is abnormal.
4. Replace U800

Maintenance Case 1:

Fault phenomenon: reverse interpolation is not recognized. Faulty component: U600

Maintenance analysis: The failure of sending repair is that it can't be charged quickly, and sometimes it enters fastboot mode. CIT test finds that the reverse insertion of USB is not recognized, and USB_CC2 is found to be short-circuited. Replace U600 for repair.

3.15 Fingerprint identification function failure

There are two different suppliers of fingerprints of Xiaomi Note3, which are displayed at CIT fingerprint sensor. If it is Goodix, the fingerprint does not need to be calibrated; If it is FPC, fingerprint needs to be calibrated. Please refer to Xiaomi 6 for specific calibration method. Please search for "Xiaomi Note3 Fingerprint Calibration File" in the knowledge base for fingerprint calibration file.

Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether the impedance of J701 to ground is normal.
3. Measure whether the voltage is normal.
4. Replace U800.